1-【初级版】学生体质信息管理系统

一. ##数据库设计

1.1 概念结构设计

实体关系如下：

- 一个学生对应多条体测记录。
- 每条体测记录只属于一个学生。

关系表示：

- `StuInfo (1) ---- (N) StuTestRecord`

1.2 逻辑结构设计

1.2.1 学生信息表 `stu_info`

表用途：
存储学生基础档案。

字段设计：

字段名	类型	约束	说明
id	BIGINT	PK, AUTO_INCREMENT	主键
stu_no	VARCHAR(30)	NOT NULL, UNIQUE	学号
stu_name	VARCHAR(50)	NOT NULL	姓名
gender	VARCHAR(10)		性别
age	INT		年龄
class_name	VARCHAR(50)		班级
height	DOUBLE		身高(cm)
weight	DOUBLE		体重(kg)
score	DOUBLE		当前体质成绩

对应代码：

- 实体：`src/main/java/com/ssm/po/StuInfo.java`
- Mapper：`src/main/resources/mybatis/StuInfoMapper.xml`

1.2.2 体测记录表 `stu_test_record`

表用途：
存储学生某一学期的体测记录，用于趋势分析和健康预警。

建议字段设计：

字段名	类型	约束	说明
id	BIGINT	PK, AUTO_INCREMENT	主键
stu_no	VARCHAR(30)	NOT NULL	学号
term	VARCHAR(30)	NOT NULL	学期
height	DOUBLE		身高(cm)
weight	DOUBLE		体重(kg)
bmi	DOUBLE		系统自动计算的 BMI
score_total	DOUBLE		该学期体测总分
warning_tag	VARCHAR(100)		预警标签
created_at	DATETIME		创建时间
updated_at	DATETIME		更新时间

关键约束：

- 唯一约束：`UNIQUE(stu_no, term)`
- 可选外键：`stu_no` 关联 `stu_info.stu_no`

对应代码：

- 实体：`src/main/java/com/ssm/po/StuTestRecord.java`
- Mapper：`src/main/resources/mybatis/StuTestRecordMapper.xml`
- 唯一更新逻辑：MyBatis `ON DUPLICATE KEY UPDATE`

1.3 建库脚本

以下脚本与当前代码实现相匹配：

```sql
DROP DATABASE IF EXISTS student;
CREATE DATABASE student DEFAULT CHARACTER SET utf8mb4;
USE student;

DROP TABLE IF EXISTS stu_test_record;
DROP TABLE IF EXISTS stu_info;

CREATE TABLE stu_info (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
```

```sql
  stu_no VARCHAR(30) NOT NULL UNIQUE,
  stu_name VARCHAR(50) NOT NULL,
  gender VARCHAR(10),
  age INT,
  class_name VARCHAR(50),
  height DOUBLE,
  weight DOUBLE,
  score DOUBLE
);

CREATE TABLE stu_test_record (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  stu_no VARCHAR(30) NOT NULL,
  term VARCHAR(30) NOT NULL,
  height DOUBLE,
  weight DOUBLE,
  bmi DOUBLE,
  score_total DOUBLE,
  warning_tag VARCHAR(100),
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  UNIQUE KEY uk_stu_term (stu_no, term)
);

-- 插入学生基础信息表 5 条数据
INSERT INTO stu_info (stu_no, stu_name, gender, age, class_name, height, weight, score)
VALUES
('S2025001', '张三', '男', 18, '高三1班', 175.5, 68.2, 85.5),
('S2025002', '李四', '女', 17, '高三1班', 162.3, 52.1, 92.0),
('S2025003', '王五', '男', 18, '高三2班', 180.0, 75.3, 78.5),
('S2025004', '赵六', '女', 17, '高三2班', 158.6, 48.7, 89.5),
('S2025005', '钱七', '男', 18, '高三1班', 172.8, 65.9, 81.0);

-- 插入学生体测记录表 5 条数据
INSERT INTO stu_test_record (stu_no, term, height, weight, bmi, score_total, warning_tag)
VALUES
('S2025001', '2024-2025学年上', 175.5, 68.2, 22.1, 88.0, '无'),
('S2025002', '2024-2025学年上', 162.3, 52.1, 19.8, 95.0, '无'),
('S2025003', '2024-2025学年上', 180.0, 75.3, 23.2, 82.5, '体重偏高'),
('S2025004', '2024-2025学年上', 158.6, 48.7, 19.3, 91.0, '无'),
('S2025005', '2024-2025学年上', 172.8, 65.9, 22.0, 85.5, '无');
```

1.4 数据库设计说明

1. `stu_info` 保存的是学生当前基础状态，适合快速查询和维护。
2. `stu_test_record` 保存的是历史体测过程数据，适合做趋势分析。
3. `stu_no` 是业务主键，贯穿前端、后端和数据库。
4. `term` 作为学期维度，保证体测数据可按时间轴组织。
5. `warning_tag` 把业务规则结果落库存储，便于后续查询、统计和页面展示。

二. ##架构思想设计

本项目采用经典的 SSM 三层架构，遵循“表现层、业务层、持久层”分离的设计思想。

架构分层如下：

1. 表现层（JSP / Vue）:主要负责前台jsp页面的表示,与控制层（controller层）关系紧密。
2. 控制层：（contoller/Action）：控制界面跳转, 负责前后端交互，接收前端发送的请求，然后调用service层，service层再返回数据给它，它再返回给前端。
3. 业务层（Service）:写主要的业务逻辑，调用DAO层，操作数据库，再对返回的数据进行各种业务上的处理，再返回给控制层。实现解耦合目的。
4. 持久层（DAO / MyBatis Mapper）：也叫数据访问层，实现对数据库的访问。完成增删改查功能，把数据返回给service层。
5. 数据层（po或model）：数据库实体层，存放实体类，实现get、set方法。属性要和数据库的一样。

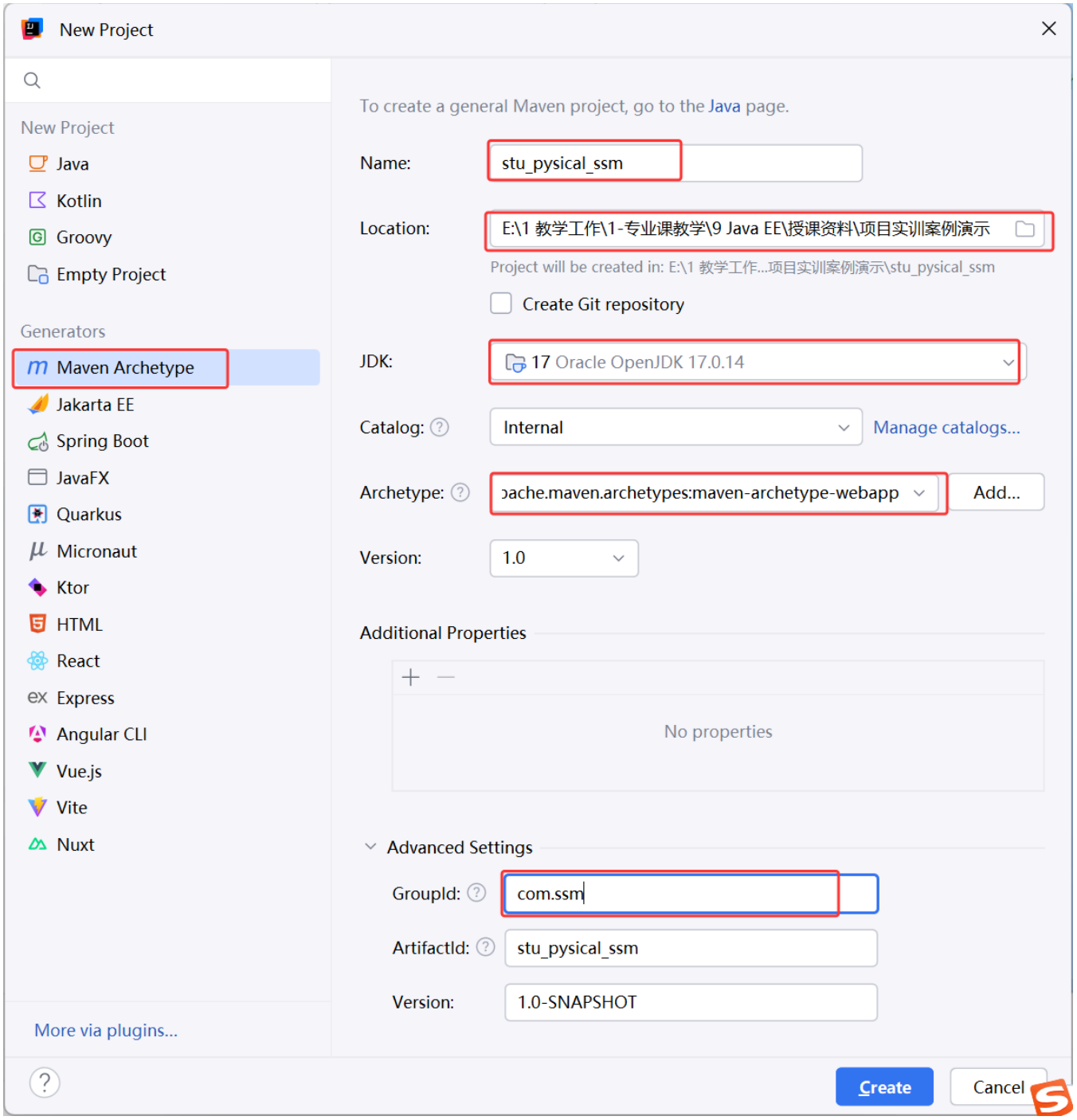
三. ##项目搭建

3.1 创建项目

1、创建Maven_Archetype项目，起名stu_pysical_ssm。找到并选择 **Archetype**：

```
org.apache.maven.archetypes:maven-archetype-webapp
```

点击 **Create



3.2 添加依赖

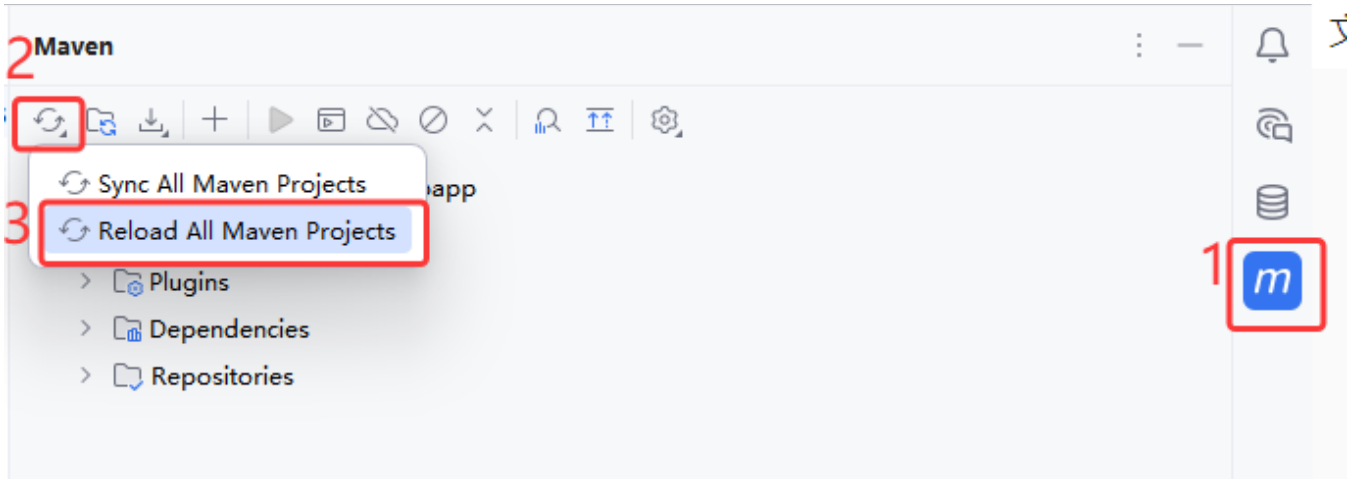
1、pom.xml中导入依赖：

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.ssm</groupId>
    <artifactId>student-physical-ssm</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>war</packaging>
```

```
<properties>
    <maven.compiler.source>8</maven.compiler.source>
    <maven.compiler.target>8</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <spring.version>5.3.34</spring.version>
    <mybatis.version>3.5.16</mybatis.version>
    <mybatis.spring.version>2.1.2</mybatis.spring.version>
</properties>

<dependencies>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>${spring.version}</version>
    </dependency>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-webmvc</artifactId>
        <version>${spring.version}</version>
    </dependency>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-jdbc</artifactId>
        <version>${spring.version}</version>
    </dependency>
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-tx</artifactId>
        <version>${spring.version}</version>
    </dependency>

    <dependency>
        <groupId>org.mybatis</groupId>
        <artifactId>mybatis</artifactId>
        <version>${mybatis.version}</version>
    </dependency>
    <dependency>
        <groupId>org.mybatis</groupId>
        <artifactId>mybatis-spring</artifactId>
        <version>${mybatis.spring.version}</version>
    </dependency>

    <dependency>
        <groupId>mysql</groupId>
        <artifactId>mysql-connector-java</artifactId>
        <version>8.0.33</version>
    </dependency>
    <dependency>
        <groupId>org.apache.commons</groupId>
        <artifactId>commons-dbcp2</artifactId>
        <version>2.12.0</version>
    </dependency>
```

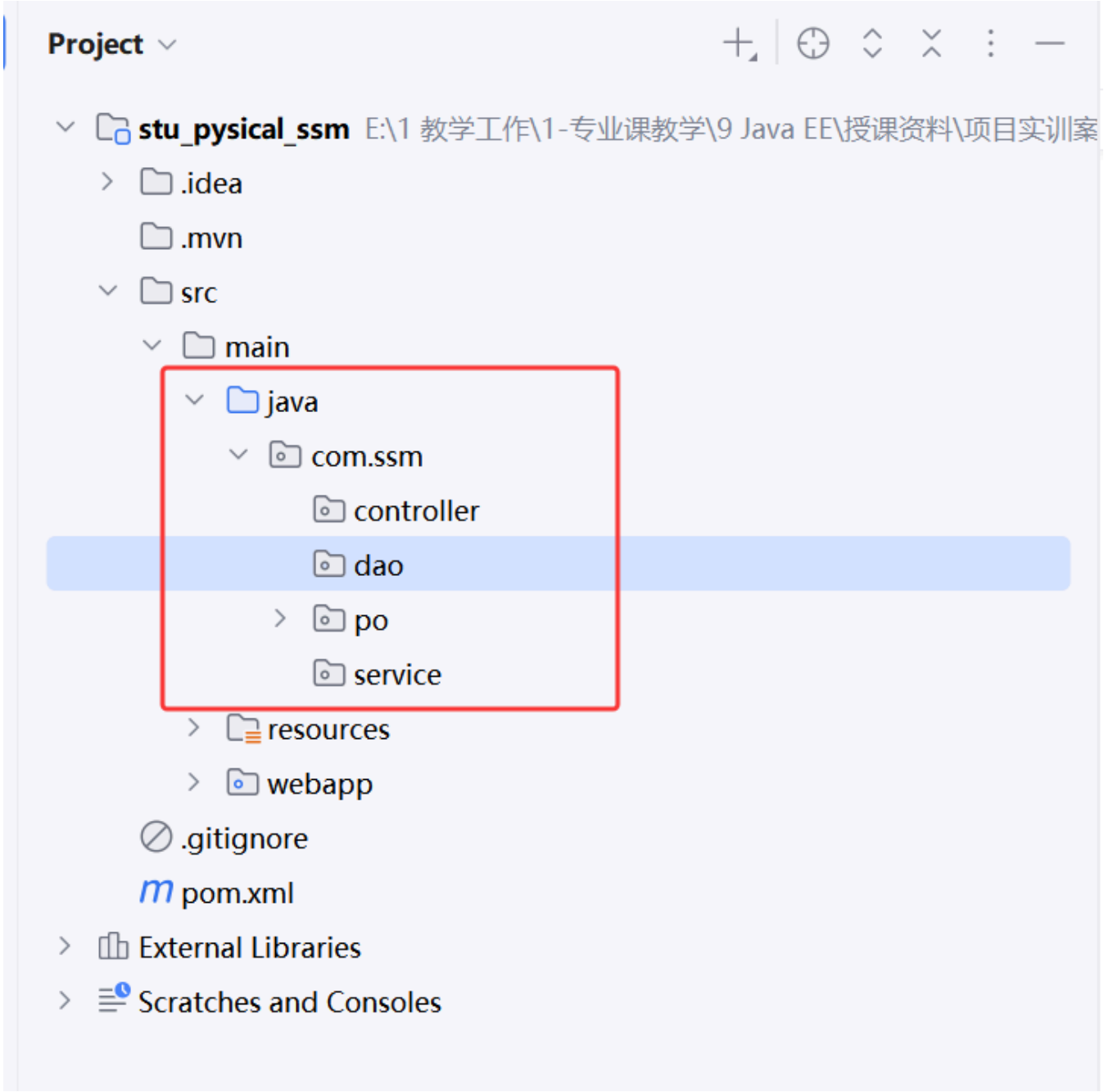
```
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>javax.servlet-api</artifactId>
            <version>4.0.1</version>
            <scope>provided</scope>
        </dependency>
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>jstl</artifactId>
            <version>1.2</version>
        </dependency>

        <dependency>
            <groupId>log4j</groupId>
            <artifactId>log4j</artifactId>
            <version>1.2.17</version>
        </dependency>
    </dependencies>

    <build>
        <finalName>student-physical-ssm</finalName>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
                <configuration>
                    <source>${maven.compiler.source}</source>
                    <target>${maven.compiler.target}</target>
                    <encoding>${project.build.sourceEncoding}</encoding>
                </configuration>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-war-plugin</artifactId>
                <version>3.4.0</version>
            </plugin>
        </plugins>
    </build>
</project>
```

2、导入依赖包。



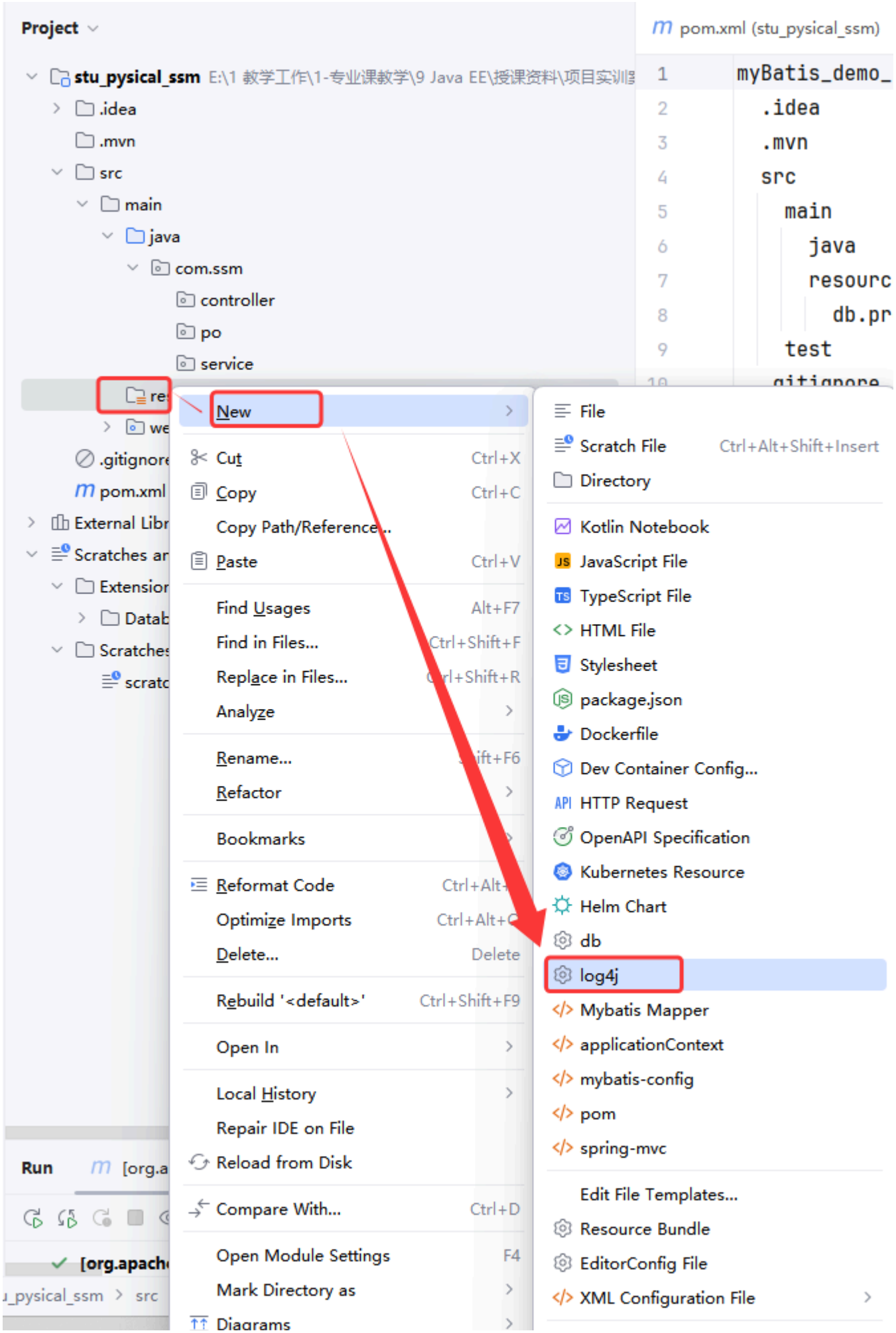
3.3 创建项目架构

1、在main文件夹中创建文件夹java，在java文件夹中创建com.ssm包，在com.ssm包中创建po\service\controller\dao四个包。



3.4 创建项目配置文件

1、在resources目录下，新建log4j.properties配置文件。


Project
stu_pysical_ssm E:\1 教学工作\1-专业课教学\9 Java EE\授课资料\项目实训
.idea
.mvn
src
main
java
com.ssm
controller
po
service
.gitignore
pom.xml
External Libr
Scratches ar
Extensior
Datab
Scratches
scratc
Run
[org.a
[org.apach
pysical_ssm
src
pom.xml (stu_pysical_ssm)
myBatis_demo_
.idea
.mvn
src
main
java
resourc
db.pr
test
New
Cut Ctrl+X
Copy Ctrl+C
Copy Path/Reference...
Paste Ctrl+V
Find Usages Alt+F7
Find in Files... Ctrl+Shift+F
Replace in Files...
Analyze
Rename...
Refactor
Bookmarks
Reformat Code
Optimize Imports
Delete... Delete
Rebuild '<default>' Ctrl+Shift+F9
Open In
Local History
Repair IDE on File
Reload from Disk
Compare With... Ctrl+D
Open Module Settings F4
Mark Directory as
Diagrams
File
Scratch File Ctrl+Alt+Shift+Insert
Directory
Kotlin Notebook
JavaScript File
TypeScript File
HTML File
Stylesheet
package.json
Dockerfile
Dev Container Config...
HTTP Request
OpenAPI Specification
Kubernetes Resource
Helm Chart
db
log4j
Mybatis Mapper
applicationContext
mybatis-config
pom
spring-mvc
Edit File Templates...
Resource Bundle
EditorConfig File
XML Configuration File

文件内容如下：

```
# Global logging configuration
log4j.rootLogger=ERROR, stdout
# MyBatis logging configuration...
log4j.logger.com.ssm=DEBUG
# Console output...
log4j.appender.stdout=org.apache.log4j.ConsoleAppender
log4j.appender.stdout.layout=org.apache.log4j.PatternLayout
log4j.appender.stdout.layout.ConversionPattern=%5p [%t] - %m%n
```

2、在resources目录下，创建db.properties配置文件

```
jdbc.driver=com.mysql.cj.jdbc.Driver
jdbc.url=jdbc:mysql://localhost:3306/student?
serverTimezone=UTC&useUnicode=true&characterEncoding=UTF-8
jdbc.username=root
jdbc.password=123456
jdbc.maxTotal=30
jdbc.maxIdle=10
jdbc.initialSize=5
```

3、在resources目录下，创建mybatis-config.xml配置文件

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE configuration
        PUBLIC "-//mybatis.org//DTD Config 3.0//EN"
        "http://mybatis.org/dtd/mybatis-3-config.dtd">
<configuration>
    <typeAliases>
        <package name="com.ssm.po"/>
    </typeAliases>
</configuration>
```

4、在resources目录下，创建applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xmlns:context="http://www.springframework.org/schema/context"
       xmlns:tx="http://www.springframework.org/schema/tx"
       xsi:schemaLocation="
       http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd
       http://www.springframework.org/schema/context
http://www.springframework.org/schema/context/spring-context.xsd
       http://www.springframework.org/schema/tx
http://www.springframework.org/schema/tx/spring-tx.xsd">

    <context:property-placeholder location="classpath:db.properties"/>
```

```
    <bean id="dataSource" class="org.apache.commons.dbcp2.BasicDataSource" destroy-
method="close">
        <property name="driverClassName" value="${jdbc.driver}"/>
        <property name="url" value="${jdbc.url}"/>
        <property name="username" value="${jdbc.username}"/>
        <property name="password" value="${jdbc.password}"/>
        <property name="maxTotal" value="${jdbc.maxTotal}"/>
        <property name="maxIdle" value="${jdbc.maxIdle}"/>
        <property name="initialSize" value="${jdbc.initialSize}"/>
    </bean>
</beans>
```

4、在resources目录下，创建spring-mvc.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:context="http://www.springframework.org/schema/context"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       http://www.springframework.org/schema/beans/spring-beans.xsd
       http://www.springframework.org/schema/context
       http://www.springframework.org/schema/context/spring-context.xsd">
    <!-- 配置 Spring MVC 要扫描的包 -->
    <context:component-scan base-package="com.ssm.controller"/>
    <!-- 配置视图解析器 -->
    <bean class="org.springframework.web.servlet.view.InternalResourceViewResolver">
        <property name="prefix" value="/"/>
        <property name="suffix" value=".jsp"/>
    </bean>
</beans>
```

四. ##功能实现

4.1 实体层po

右击po包new——> Java class，命名为StuInfo

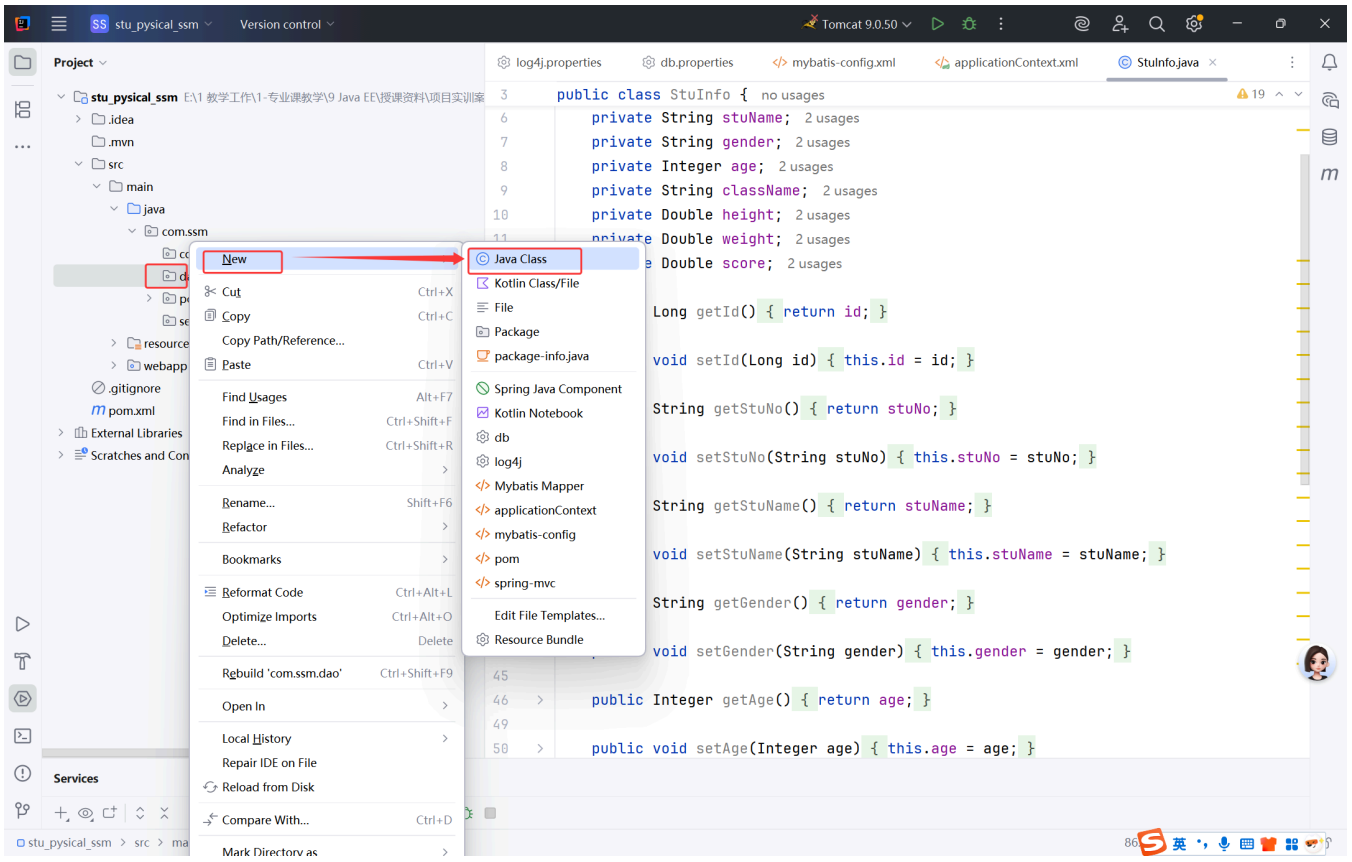
```
public class StuInfo {
    private Long id;
    private String stuNo;
    private String stuName;
    private String gender;
    private Integer age;
    private String className;
    private Double height;
    private Double weight;
    private Double score;

    public Long getId() {
        return id;
    }
```

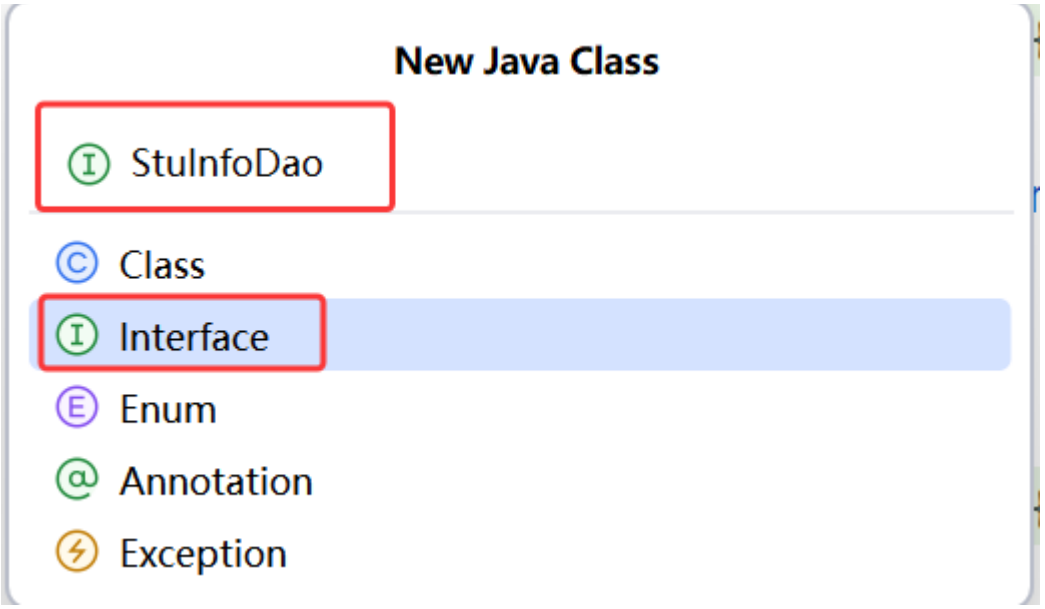
```java
    public void setId(Long id) {
        this.id = id;
    }

    public String getStuNo() {
        return stuNo;
    }

    public void setStuNo(String stuNo) {
        this.stuNo = stuNo;
    }

    public String getStuName() {
        return stuName;
    }

    public void setStuName(String stuName) {
        this.stuName = stuName;
    }

    public String getGender() {
        return gender;
    }

    public void setGender(String gender) {
        this.gender = gender;
    }

    public Integer getAge() {
        return age;
    }

    public void setAge(Integer age) {
        this.age = age;
    }

    public String getClassName() {
        return className;
    }

    public void setClassName(String className) {
        this.className = className;
    }

    public Double getHeight() {
        return height;
    }

    public void setHeight(Double height) {
        this.height = height;
    }
```

```
    public Double getWeight() {
        return weight;
    }

    public void setWeight(Double weight) {
        this.weight = weight;
    }

    public Double getScore() {
        return score;
    }

    public void setScore(Double score) {
        this.score = score;
    }
}
```

4.2 数据访问层dao

1、创建数据访问层接口StuInfoDao






```java
public interface StuInfoDao {
    //    新增学生信息
    int insert(StuInfo stuInfo);
    //    删除学生信息
    int deleteByStuNo(@Param("stuNo") String stuNo);
    //    更新学生信息
    int updateByStuNo(StuInfo stuInfo);

    //    学生信息查询（通过学号、关键字查询以及查询全部）
    StuInfo findByStuNo(@Param("stuNo") String stuNo);
    List<StuInfo> findByKeyword(@Param("keyword") String keyword);
    List<StuInfo> findAll();
}
```

2、实现Mapper文件。在resources目录中，创建mybatis目录文件夹，在mybatis目录下创建StuInfoMapper.xml文件

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE mapper
        PUBLIC "-//mybatis.org//DTD Mapper 3.0//EN"
        "http://mybatis.org/dtd/mybatis-3-mapper.dtd">

<!-- namespace 对应你的 Mapper 接口全类名 -->
<mapper namespace="com.ssm.dao.StuInfoDao">
    <resultMap id="StuInfoMap" type="com.ssm.po.StuInfo">
        <id property="id" column="id"/>
        <result property="stuNo" column="stu_no"/>
        <result property="stuName" column="stu_name"/>
        <result property="gender" column="gender"/>
        <result property="age" column="age"/>
        <result property="className" column="class_name"/>
        <result property="height" column="height"/>
        <result property="weight" column="weight"/>
        <result property="score" column="score"/>
    </resultMap>
```

```
    <insert id="insert" parameterType="com.ssm.po.StuInfo" useGeneratedKeys="true"
keyProperty="id">
        INSERT INTO stu_info(stu_no, stu_name, gender, age, class_name, height, weight,
score)
        VALUES(#{stuNo}, #{stuName}, #{gender}, #{age}, #{className}, #{height}, #{weight},
#{score})
    </insert>

    <delete id="deleteByStuNo">
        DELETE FROM stu_info WHERE stu_no = #{stuNo}
    </delete>

    <update id="updateByStuNo" parameterType="com.ssm.po.StuInfo">
        UPDATE stu_info
        SET stu_name = #{stuName},
            gender = #{gender},
            age = #{age},
            class_name = #{className},
            height = #{height},
            weight = #{weight},
            score = #{score}
        WHERE stu_no = #{stuNo}
    </update>

    <select id="findByStuNo" resultMap="StuInfoMap">
        SELECT id, stu_no, stu_name, gender, age, class_name, height, weight, score
        FROM stu_info
        WHERE stu_no = #{stuNo}
    </select>

    <select id="findByKeyword" resultMap="StuInfoMap">
        SELECT id, stu_no, stu_name, gender, age, class_name, height, weight, score
        FROM stu_info
        WHERE stu_no LIKE CONCAT('%', #{keyword}, '%')
           OR stu_name LIKE CONCAT('%', #{keyword}, '%')
        ORDER BY id DESC
    </select>

    <select id="findAll" resultMap="StuInfoMap">
        SELECT id, stu_no, stu_name, gender, age, class_name, height, weight, score
        FROM stu_info
        ORDER BY id DESC
    </select>
</mapper>
```

4.3 业务层service

1、首先实现service接口类StuInfoService

```java
public interface StuInfoService {
    boolean save(StuInfo stuInfo);

    boolean removeByStuNo(String stuNo);

    boolean updateByStuNo(StuInfo stuInfo);

    StuInfo findByStuNo(String stuNo);

    List<StuInfo> findByKeyword(String keyword);

    List<StuInfo> findAll();
}
```

2、右击service包，新建包名impl，右击impl包new——>java class，命名为“StuInfoServiceImpl”

```java
@Service
public class StuInfoServiceImpl implements StuInfoService {
    @Autowired
    private StuInfoDao stuInfoDao;

    @Override
    public boolean save(StuInfo stuInfo) {
        return stuInfoDao.insert(stuInfo) > 0;
    }

    @Override
    public boolean removeByStuNo(String stuNo) {
        return stuInfoDao.deleteByStuNo(stuNo) > 0;
    }

    @Override
    public boolean updateByStuNo(StuInfo stuInfo) {
        return stuInfoDao.updateByStuNo(stuInfo) > 0;
    }

    @Override
    public StuInfo findByStuNo(String stuNo) {
        return stuInfoDao.findByStuNo(stuNo);
    }

    @Override
    public List<StuInfo> findByKeyword(String keyword) {
        if (keyword == null || keyword.trim().isEmpty()) {
            return stuInfoDao.findAll();
        }
        return stuInfoDao.findByKeyword(keyword.trim());
    }

    @Override
    public List<StuInfo> findAll() {
        return stuInfoDao.findAll();
```

```
    }
}
```

4.4 控制层controller

在controller目录中，创建类StuInfoController。编写增加学生信息功能。

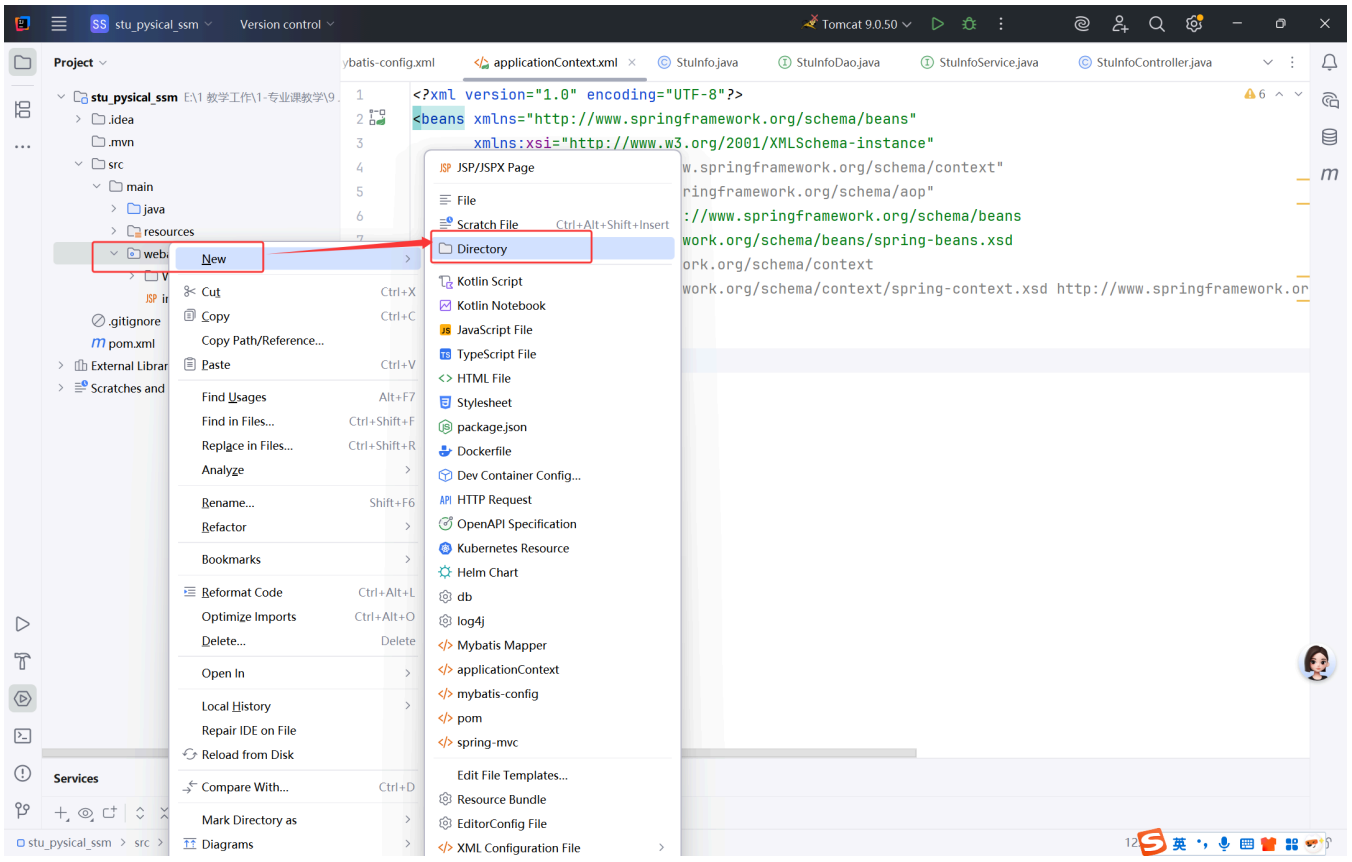
```
@Controller
@RequestMapping("/stu")
public class StuInfoController {

    @Autowired
    private StuInfoService stuInfoService;

    @GetMapping("/insert")
    public String insertPage() {
        return "page/insert-stu";
    }

    @PostMapping("/insert")
    public String insert(StuInfo stuInfo, Model model) {
        boolean ok = stuInfoService.save(stuInfo);
        model.addAttribute("ok", ok);
        model.addAttribute("message", ok ? "新增成功" : "新增失败");
        return "insert-stu-tijiao";
    }
}
```

4.5 表示层（基于JSP）

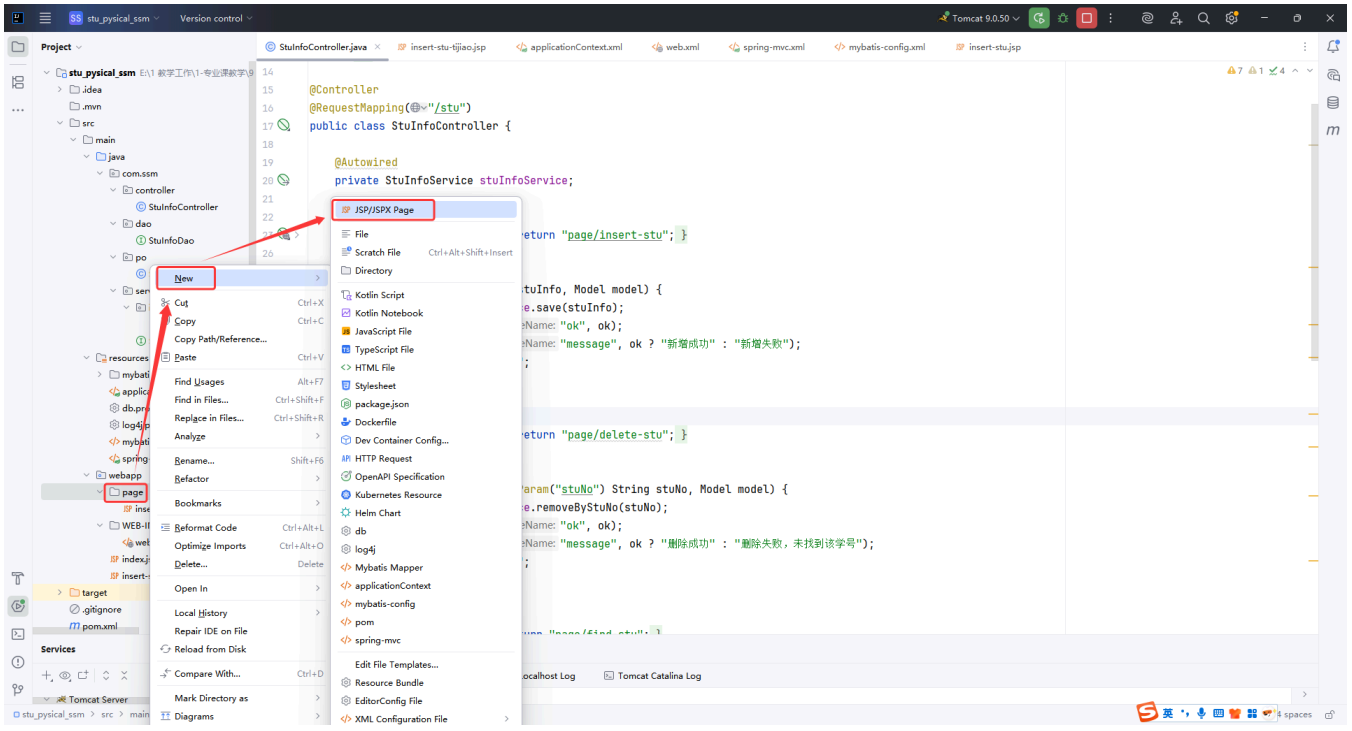
1、修改index.jsp页面

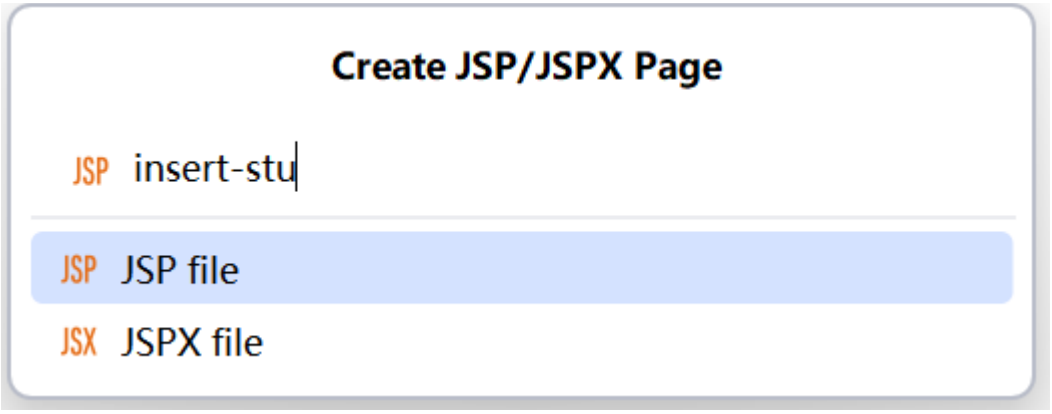
```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>学生体质信息管理系统</title>
    <style>
        body { font-family: "Microsoft YaHei", sans-serif; padding: 30px; }
        a { display: inline-block; margin: 8px 12px 8px 0; }
    </style>
</head>
<body>
<h2>学生体质信息管理系统（SSM）</h2>
<p>请选择功能：</p>
<a href="${pageContext.request.contextPath}/stu/insert">新增学生体质信息</a>
<a href="${pageContext.request.contextPath}/stu/delete">删除学生体质信息</a>
<a href="${pageContext.request.contextPath}/stu/find">查询学生体质信息</a>
<a href="${pageContext.request.contextPath}/stu/update">修改学生体质信息</a>
</body>
</html>
```

2、在webapp目录下，创建目录，命名page



3、在page目录下，创建页面insert-stu






```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head><title>新增学生体质信息</title></head>
<body>
<h3>新增学生体质信息</h3>
<form action="${pageContext.request.contextPath}/stu/insert" method="post">
    学号：<input name="stuNo" required/><br/>
    姓名：<input name="stuName" required/><br/>
    性别：<input name="gender"/><br/>
    年龄：<input name="age" type="number"/><br/>
    班级：<input name="className"/><br/>
    身高(cm)：<input name="height" type="number" step="0.1"/><br/>
    体重(kg)：<input name="weight" type="number" step="0.1"/><br/>
    成绩：<input name="score" type="number" step="0.1"/><br/>
    <button type="submit">提交</button>
</form>
<a href="${pageContext.request.contextPath}/">返回首页</a>
</body>
</html>
```

4、在webapp目录下，创建jsp页面insert-stu-tijiao

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head><title>新增提交结果</title></head>
<body>
<h3>${message}</h3>
<a href="${pageContext.request.contextPath}/stu/insert">继续新增</a>
<a href="${pageContext.request.contextPath}/">返回首页</a>
</body>
</html>
```

五. ##SSM整合

1、修改applicationContext.xml文件

```
<?xml version="1.0" encoding="UTF-8"?>
```

```xml
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xmlns:context="http://www.springframework.org/schema/context"
       xmlns:tx="http://www.springframework.org/schema/tx"
       xsi:schemaLocation="
       http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd
       http://www.springframework.org/schema/context
http://www.springframework.org/schema/context/spring-context.xsd
       http://www.springframework.org/schema/tx
http://www.springframework.org/schema/tx/spring-tx.xsd">

<!--    <context:component-scan base-package="com.ssm"/>-->
<!--
这段代码是 SSM 整合的核心配置，作用只有 3 个：
1、管理数据库连接（事务）
2、整合 MyBatis（SqlSessionFactory）
3、自动扫描 DAO/Mapper 接口，生成实现类
-->
    <context:property-placeholder location="classpath:db.properties"/>
    <bean id="dataSource" class="org.apache.commons.dbcp2.BasicDataSource" destroy-
method="close">
        <property name="driverClassName" value="${jdbc.driver}"/>
        <property name="url" value="${jdbc.url}"/>
        <property name="username" value="${jdbc.username}"/>
        <property name="password" value="${jdbc.password}"/>
        <property name="maxTotal" value="${jdbc.maxTotal}"/>
        <property name="maxIdle" value="${jdbc.maxIdle}"/>
        <property name="initialSize" value="${jdbc.initialSize}"/>
    </bean>

    <bean id="transactionManager"
class="org.springframework.jdbc.datasource.DataSourceTransactionManager">
        <property name="dataSource" ref="dataSource"/>
    </bean>

    <tx:advice id="txAdvice" transaction-manager="transactionManager">
        <tx:attributes>
            <tx:method name="find*" read-only="true"/>
            <tx:method name="*" propagation="REQUIRED"/>
        </tx:attributes>
    </tx:advice>

    <bean id="sqlSessionFactory" class="org.mybatis.spring.SqlSessionFactoryBean">
        <property name="dataSource" ref="dataSource"/>
        <property name="configLocation" value="classpath:mybatis-config.xml"/>
        <property name="mapperLocations" value="classpath:mybatis/StuInfoMapper.xml"/>
    </bean>

    <bean class="org.mybatis.spring.mapper.MapperScannerConfigurer">
        <property name="basePackage" value="com.ssm.dao"/>
    </bean>
```

```
    <context:component-scan base-package="com.ssm.service"/>

    <bean
class="org.springframework.aop.framework.autoproxy.DefaultAdvisorAutoProxyCreator"/>
    <bean class="org.springframework.aop.support.RegexpMethodPointcutAdvisor">
        <property name="advice" ref="txAdvice"/>
        <property name="patterns" value="com.ssm.service.*.*(..)"/>
    </bean>

</beans>
```

2、配置处理器映射信息和视图解析器。在resources目录下，创建Spring MVC的配置文件spring-mvc.xml。

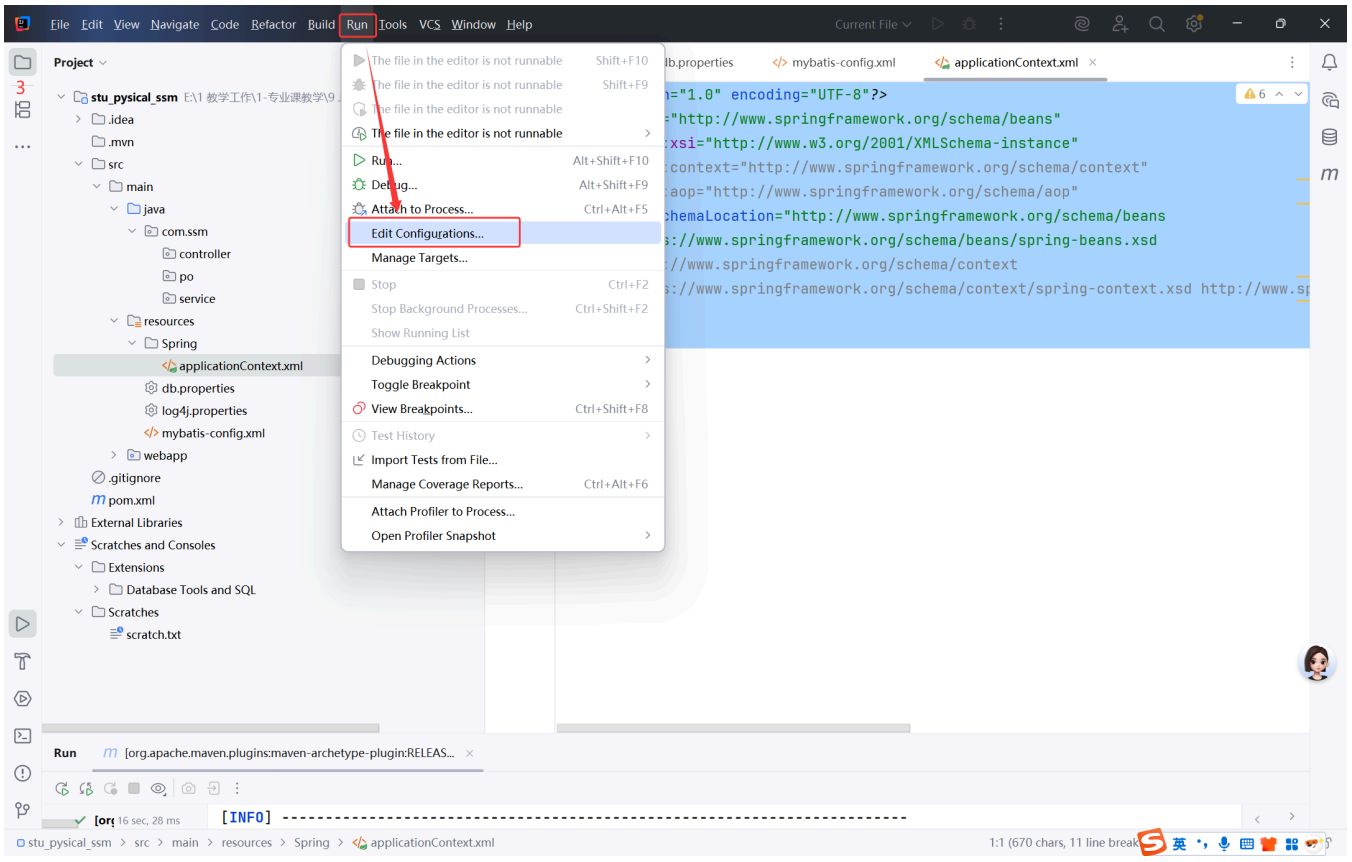
```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:context="http://www.springframework.org/schema/context"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       http://www.springframework.org/schema/beans/spring-beans.xsd
       http://www.springframework.org/schema/context
       http://www.springframework.org/schema/context/spring-context.xsd">
    <!-- 配置 Spring MVC 要扫描的包 -->
    <context:component-scan base-package="com.ssm.controller"/>
    <!-- 配置视图解析器 -->
    <bean class="org.springframework.web.servlet.view.InternalResourceViewResolver">
        <property name="prefix" value="/"/>
        <property name="suffix" value=".jsp"/>
    </bean>
</beans>
```

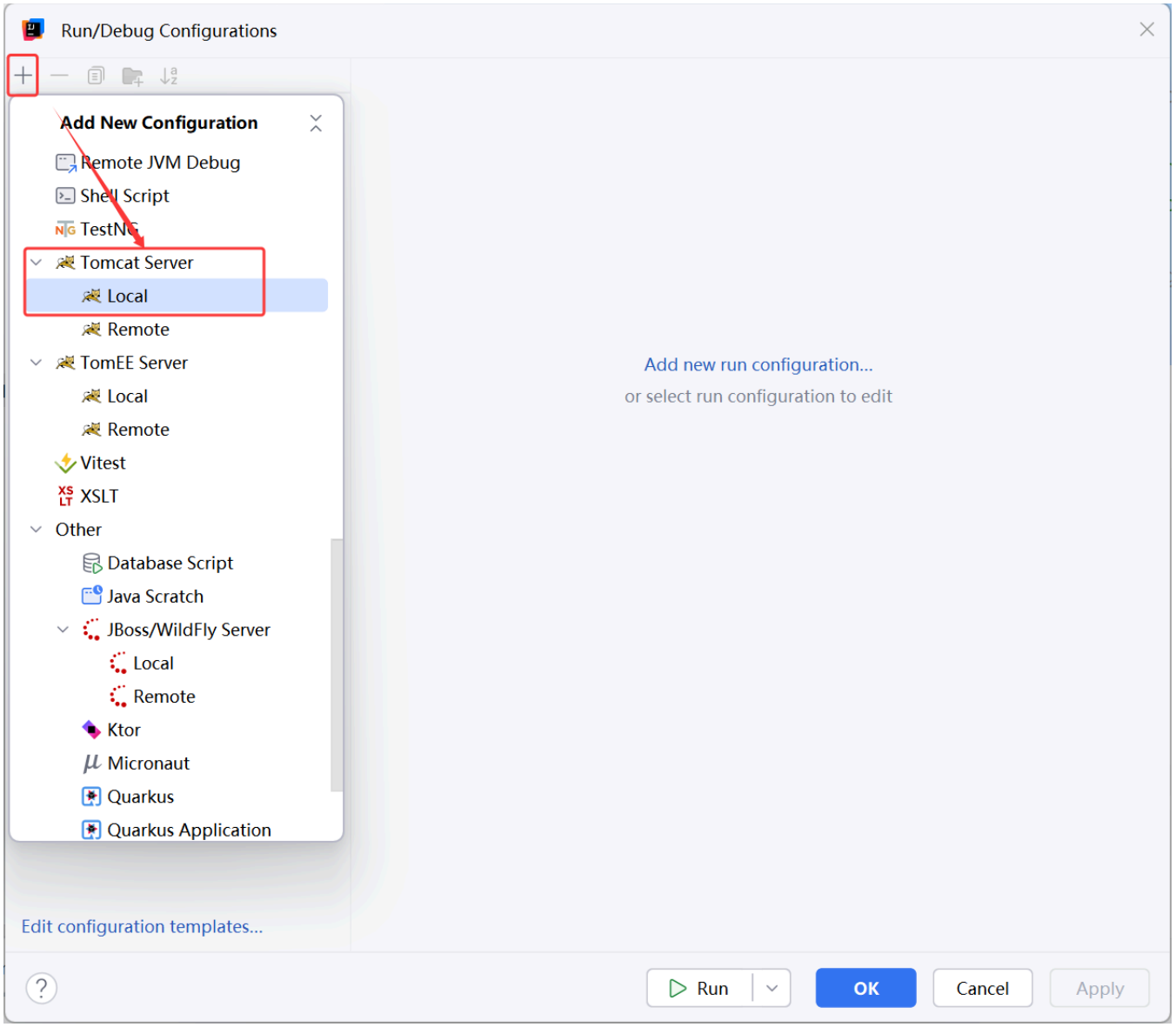
3、配置前端控制器。在项目的web.xml文件中进行Spring MVC前端控制器的配置。

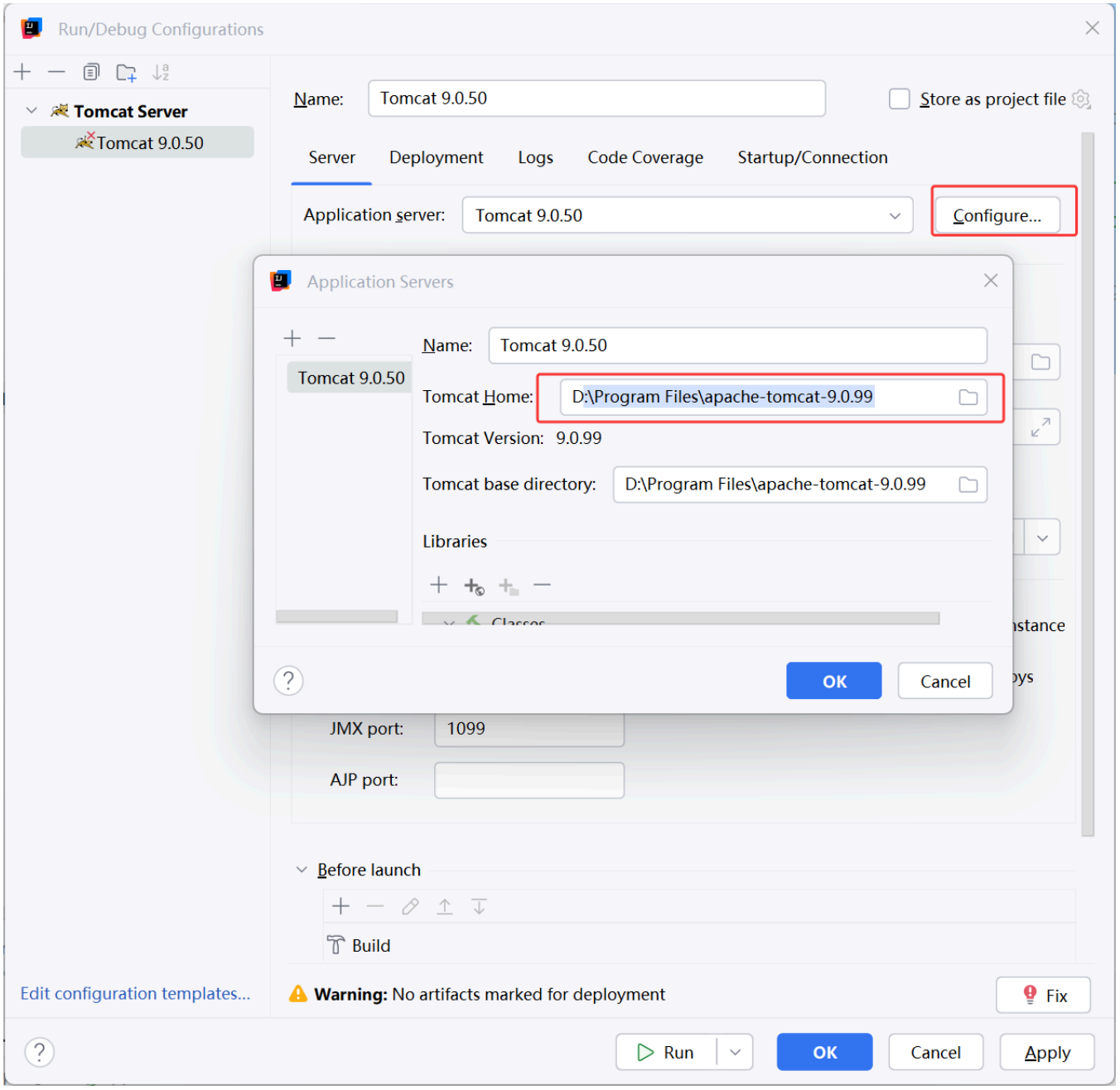
```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd"
         version="4.0">

  <display-name>student-physical-ssm</display-name>

  <context-param>
    <param-name>contextConfigLocation</param-name>
    <param-value>classpath:applicationContext.xml</param-value>
  </context-param>

  <listener>
    <listener-class>org.springframework.web.context.ContextLoaderListener</listener-class>
  </listener>

  <servlet>
```

```
    <servlet-name>springmvc</servlet-name>
    <servlet-class>org.springframework.web.servlet.DispatcherServlet</servlet-class>
    <init-param>
      <param-name>contextConfigLocation</param-name>
      <param-value>classpath:spring-mvc.xml</param-value>
    </init-param>
    <load-on-startup>1</load-on-startup>
  </servlet>

  <servlet-mapping>
    <servlet-name>springmvc</servlet-name>
    <url-pattern>/</url-pattern>
  </servlet-mapping>

  <filter>
    <filter-name>encodingFilter</filter-name>
    <filter-class>org.springframework.web.filter.CharacterEncodingFilter</filter-class>
    <init-param>
      <param-name>encoding</param-name>
      <param-value>UTF-8</param-value>
    </init-param>
    <init-param>
      <param-name>forceEncoding</param-name>
      <param-value>true</param-value>
    </init-param>
  </filter>

  <filter-mapping>
    <filter-name>encodingFilter</filter-name>
    <url-pattern>/*</url-pattern>
  </filter-mapping>

  <welcome-file-list>
    <welcome-file>index.jsp</welcome-file>
  </welcome-file-list>
</web-app>
```

六. ##配置Tomcat服务器

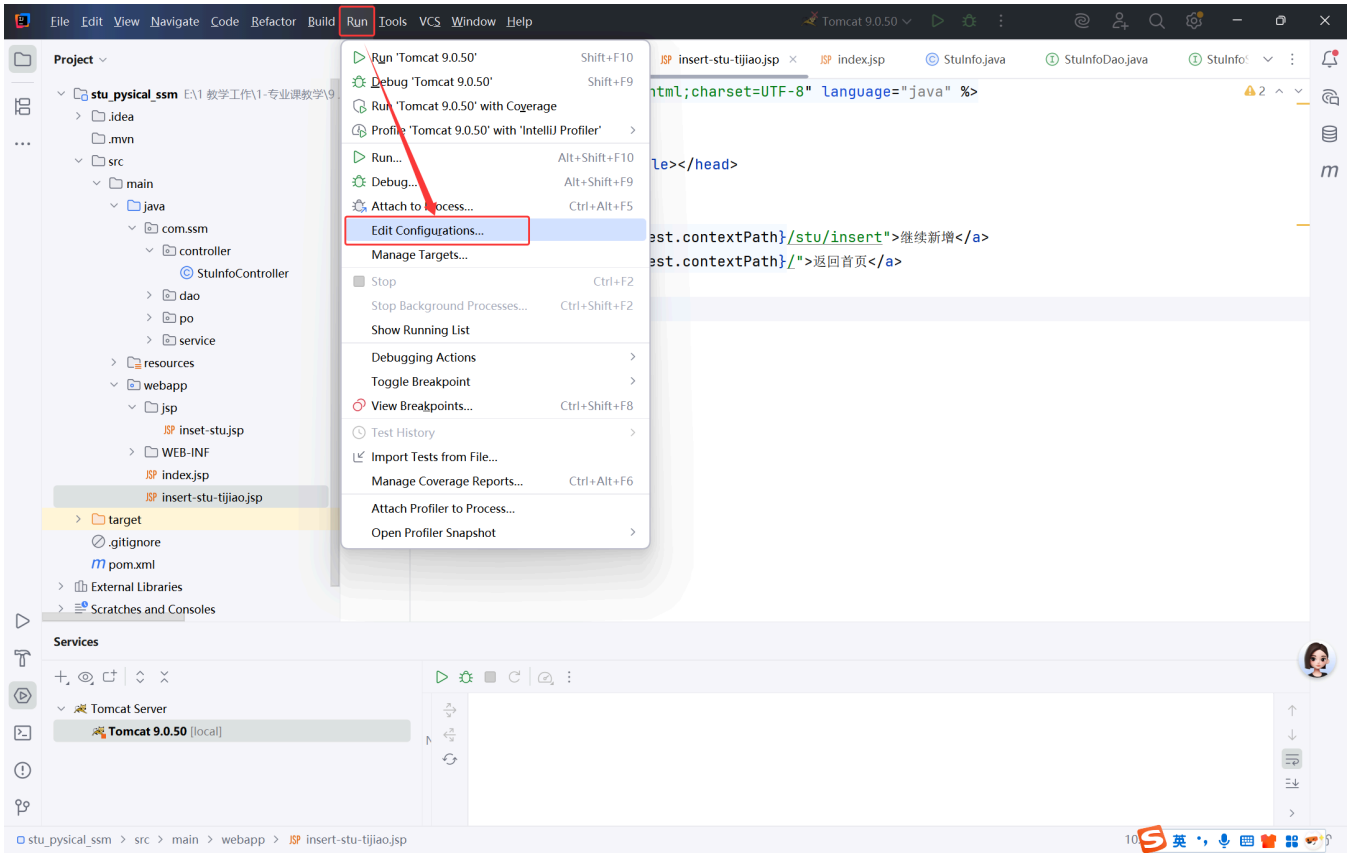



Run/Debug Configurations
Add New Configuration
Remote JVM Debug
Shell Script
TestNG
Tomcat Server
Local
Remote
TomEE Server
Local
Remote
Vitest
XSLT
Other
Database Script
Java Scratch
JBoss/WildFly Server
Local
Remote
Ktor
Micronaut
Quarkus
Quarkus Application
Edit configuration templates...
Add new run configuration...
or select run configuration to edit
Run
OK
Cancel
Apply

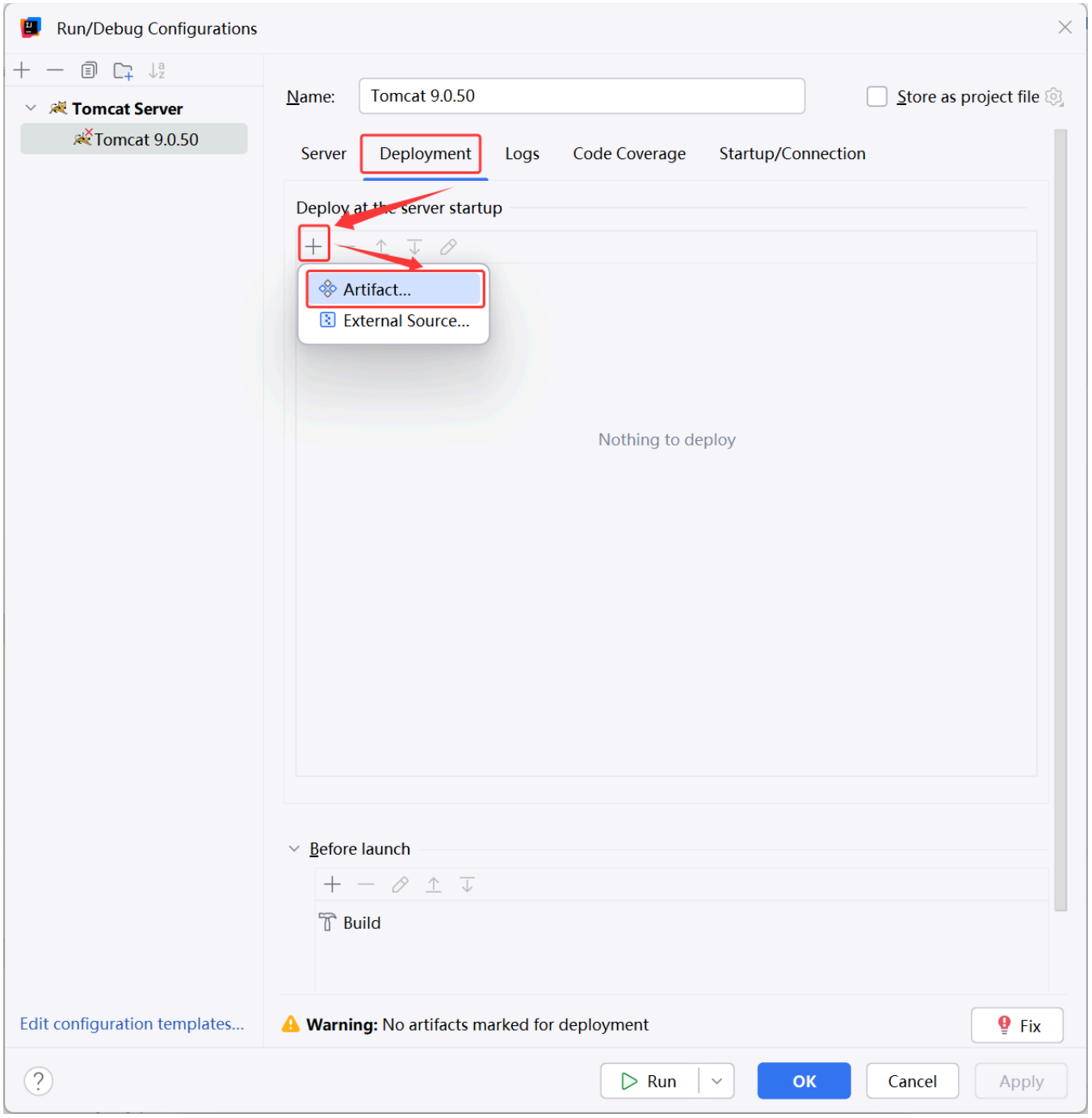


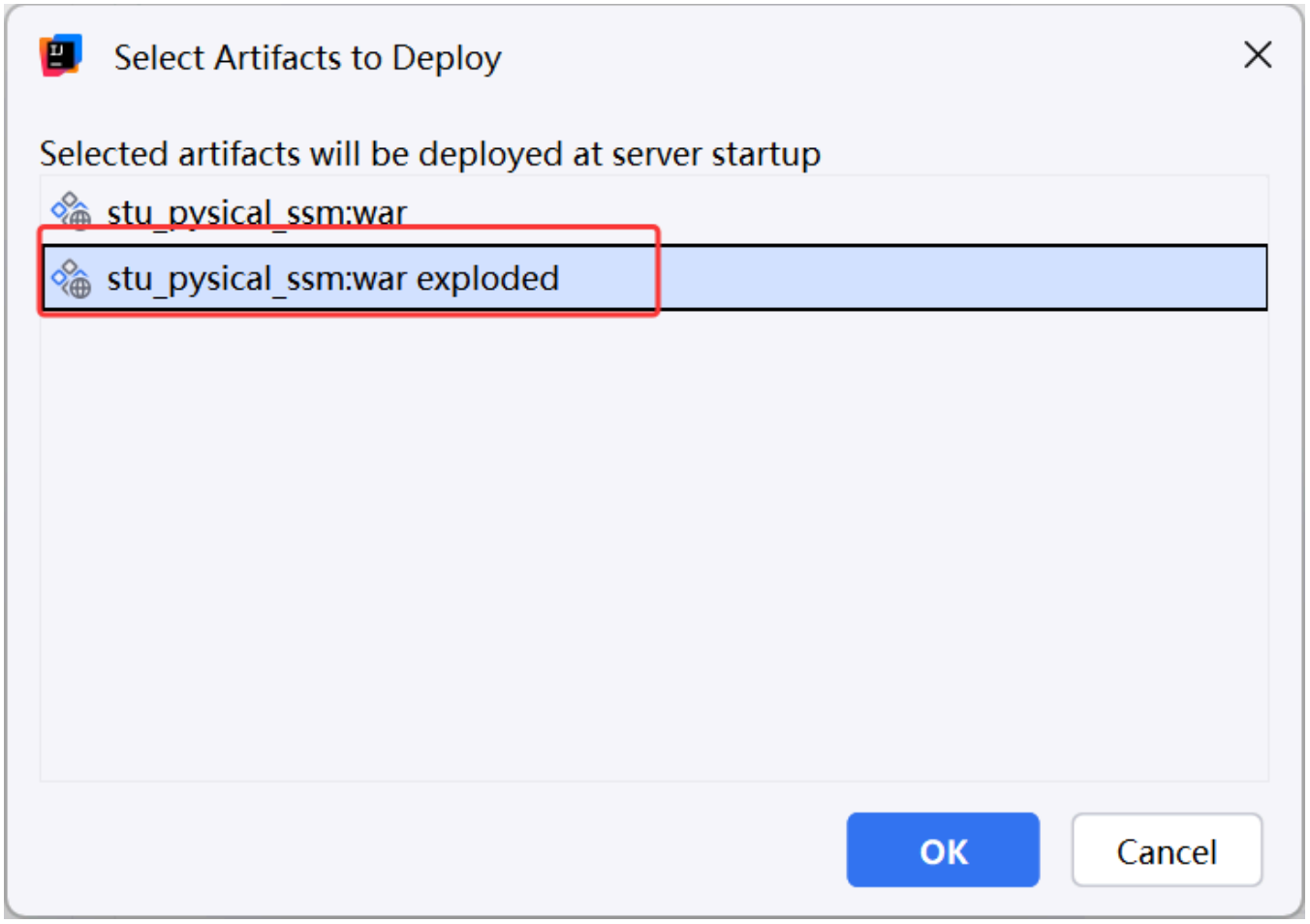
七. ##部署项目

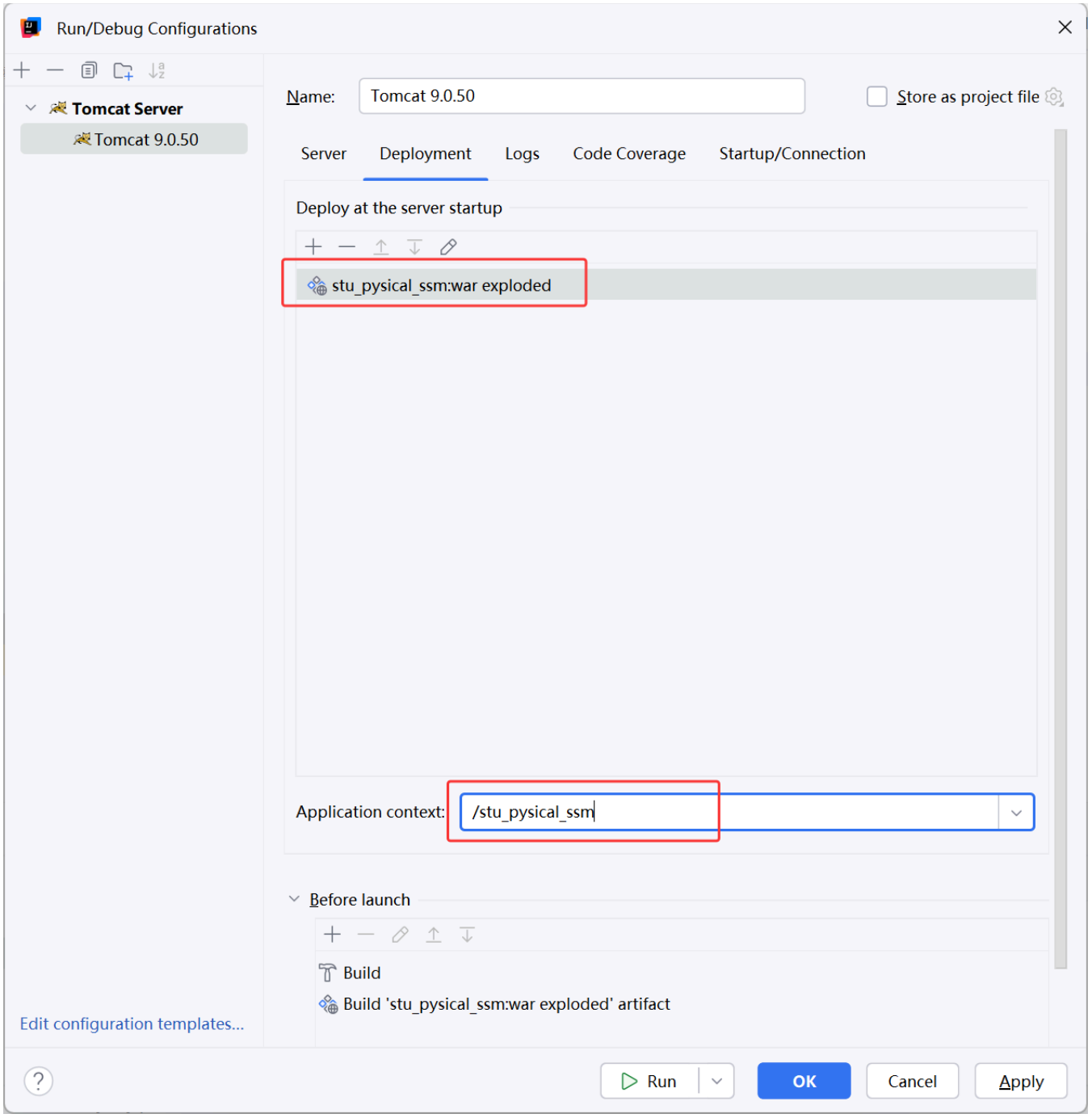
1、进入运行配置页面


File Edit View Navigate Code Refactor Build Run Tools VCS Window Help
Tomcat 9.0.50
Project
stu_pysical_ssm E:\1 教学工作\1-专业课教学\9
.idea
.mvn
src
main
java
com.ssm
controller
StuInfoController
dao
po
service
resources
webapp
jsp
inset-stu.jsp
WEB-INF
index.jsp
insert-stu-tijiao.jsp
target
.gitignore
pom.xml
External Libraries
Scratches and Consoles
Run 'Tomcat 9.0.50' Shift+F10
Debug 'Tomcat 9.0.50' Shift+F9
Run 'Tomcat 9.0.50' with Coverage
Profile 'Tomcat 9.0.50' with 'IntelliJ Profiler'
Run... Alt+Shift+F10
Debug... Alt+Shift+F9
Attach to Process... Ctrl+Alt+F5
Edit Configurations...
Manage Targets...
Stop Ctrl+F2
Stop Background Processes... Ctrl+Shift+F2
Show Running List
Debugging Actions
Toggle Breakpoint
View Breakpoints... Ctrl+Shift+F8
Test History
Import Tests from File...
Manage Coverage Reports... Ctrl+Alt+F6
Attach Profiler to Process...
Open Profiler Snapshot
insert-stu-tijiao.jsp
index.jsp
StuInfo.java
StuInfoDao.java
tml;charset="UTF-8" language="java" %>
le></head>
est.contextPath}/stu/insert">继续新增</a>
est.contextPath}/">返回首页</a>
Services
Tomcat Server
Tomcat 9.0.50 [local]
stu_pysical_ssm > src > main > webapp > insert-stu-tijiao.jsp

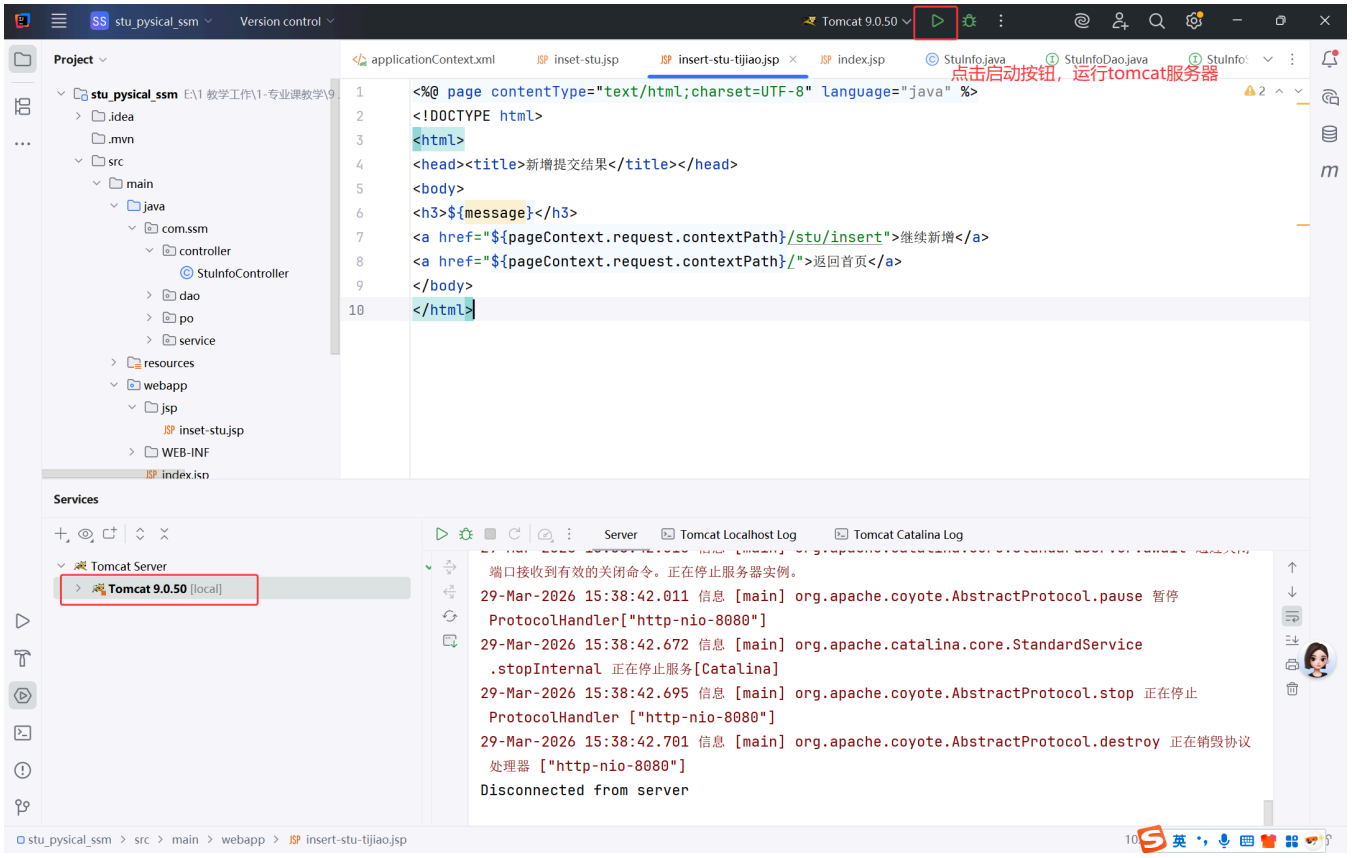
2、部署项目


Run/Debug Configurations
Tomcat Server
Tomcat 9.0.50
Name:
Tomcat 9.0.50
Store as project file
Server
Deployment
Logs
Code Coverage
Startup/Connection
Deploy at the server startup
Artifact...
External Source...
Nothing to deploy
Before launch
Build
Edit configuration templates...
Warning: No artifacts marked for deployment
Fix
Run
OK
Cancel
Apply


Select Artifacts to Deploy
Selected artifacts will be deployed at server startup
stu_pysical_ssm:war
stu_pysical_ssm:war exploded
OK
Cancel



3、点击启动按钮，运行tomcat服务器


stu_pysical_ssm
Version control
Tomcat 9.0.50
点击启动按钮，运行tomcat服务器
Project
applicationContext.xml
inset-stu.jsp
insert-stu-tijiao.jsp
index.jsp
StuInfo.java
StuInfoDao.java
stu_pysical_ssm E:\1 教学工作\1-专业课教学\9
.idea
.mvn
src
main
java
com.ssm
controller
StuInfoController
dao
po
service
resources
webapp
jsp
inset-stu.jsp
WEB-INF
index.jsp
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head><title>新增提交结果</title></head>
<body>
<h3>${message}</h3>
<a href="${pageContext.request.contextPath}/stu/insert">继续新增</a>
<a href="${pageContext.request.contextPath}/">返回首页</a>
</body>
</html>
Services
Tomcat Server
Tomcat 9.0.50 [local]
Server
Tomcat Localhost Log
Tomcat Catalina Log
端口接收到有效的关闭命令。正在停止服务器实例。
29-Mar-2026 15:38:42.011 信息 [main] org.apache.coyote.AbstractProtocol.pause 暂停 ProtocolHandler["http-nio-8080"]
29-Mar-2026 15:38:42.672 信息 [main] org.apache.catalina.core.StandardService .stopInternal 正在停止服务[Catalina]
29-Mar-2026 15:38:42.695 信息 [main] org.apache.coyote.AbstractProtocol.stop 正在停止 ProtocolHandler ["http-nio-8080"]
29-Mar-2026 15:38:42.701 信息 [main] org.apache.coyote.AbstractProtocol.destroy 正在销毁协议处理器 ["http-nio-8080"]
Disconnected from server
stu_pysical_ssm > src > main > webapp > insert-stu-tijiao.jsp

八. ##运行结果

学生体质信息管理系统
localhost:8080/stu_pysical_ssm/
学生体质信息管理系统（SSM）
请选择功能：
新增学生体质信息 删除学生体质信息 查询学生体质信息 修改学生体质信息

九. ##测试新增学生信息功能

十. ##继续完成其他功能